Akhil S L

Research Associate · Aerospace Engineering

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Research Interests

Mesh-free/ Mesh-less methods · Nonlocal elasticity · Micro/nano-structures · Computational solid and structural mechanics · Scientific computing and numerical methods

Education

Aug 2021 – present **Doctor of Philosophy, Aerospace Engineering**
Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram
Thesis: "Meshfree Method for Static and Dynamic Analysis of One-Dimensional Nonlocal Elastic Structures"
Supervisor: Dr. Praveen Krishna I R, Structural Dynamics & Vibration Laboratory, IIST

June 2019 – June 2021 **Master of Technology, Machine Design**
Government Engineering College, Thiruvananthapuram
GPA: 9.7 / 10

Aug 2013 – Nov 2017 **Bachelor of Technology, Mechanical Engineering**
Government Engineering College, Idukki
GPA: 7.6 / 10

Research Experience

Aug 2021 – present **Doctoral Researcher**
Structural Dynamics & Vibration Laboratory, IIST, Thiruvananthapuram

- Developed the Element-Free Galerkin method (EFGM) for strain-driven (Eringen), and stress-driven (Romano–Barretta) nonlocal elasticity problems in rods, Euler–Bernoulli beams, and Timoshenko beams.
- Implemented Lagrange multiplier and scaled transformation methods for essential boundary condition enforcement in MATLAB.
- Extended the framework to dynamic problems including free vibration, forced response, and wave propagation.

March 2021 **M.Tech Project**
Government Engineering College, Thiruvananthapuram
Reduced-order modelling of systems with localised nonlinearity using the Craig–Bampton method and harmonic balance method (tools: ANSYS, MATLAB).

Publications

Journal Articles

2025 Akhil S L, Krishna I R Praveen, Aswathy M. "Effect of non-dimensional length scale in element free Galerkin method for classical and strain driven nonlocal elasto-static problems." *Computers & Structures*, Vol. 312, May 2025, pp. 1–11. DOI: 10.1016/j.compstruc.2025.107724

2025 S L Akhil, I R Praveen Krishna. "Element-Free Galerkin Method for Elastostatic Analysis of Nonlocal Stress-Driven Bernoulli–Euler Beams." *ASCE Journal of Engineering Mechanics*, Vol. 151, No. 10, pp. 1–12. DOI: 10.1061/JENMDT.EMENG-8428

2026 *[Submitted for review]*S L Akhil, I R Praveen Krishna. "An exact constitutive boundary treatment and generalized stress-driven element-free Galerkin method for nanorods under static loading"

2026 *[Submitted for review]*S L Akhil, I R Praveen Krishna. "An exact constitutive boundary enforced stress-driven element-free Galerkin method for vibration and wave propagation in nanorods"

2026 *[Submitted for review]*S L Akhil, M Aswathy, I R Praveen Krishna. "A Stress-Driven Nonlocal Element-Free Galerkin Framework for the Elastodynamics of Bernoulli-Euler Nanobeams"

Conference/ Conference Proceedings

Dec 2024 Akhil S L, I R Praveen Krishna, Gaurab Kumar Khanra. "Element free Galerkin method (EFGM) for static analysis of nonlocal elastic solids using differential and integral strain driven models." In *4th International Conference on Mechanics of Advanced Materials and Structures (ICMAMS)*, 11–13 Dec 2024, Bengaluru.

Sept 2025 Akhil S L, I R Praveen Krishna, Aswathy M. "Element free Galerkin method for Timoshenko nano beams using strain driven nonlocal model." In *4th International Mechanical Engineering Congress & Exposition India (IMECE India)*, 10–13 Sept 2025, Hyderabad.

July 2026 Akhil S L, I R Praveen Krishna. "Element-Free Galerkin Formulation of an Integral Nonlocal Model with Modified Kernel for Cantilever Nano-Beams" In *International Conference on mechanics of Advanced Materials and Structures, Toulouse, France, July 1-3,2026.*

Sept 2026 Akhil S L, I R Praveen Krishna. "Numerical Solution of Free Vibration of Micro- and Nano-Beams Using the Integral Form of Eringen's Strain-Driven Model." In *International Mechanical Engineering Congress & Exposition (IMECE) India, Chennai, Sept 3-5,2026.*

Dec 2026 *[upcoming]*Akhil S L, I R Praveen Krishna. "Element-free Galerkin method for Buckling Analysis of Stress-Driven Nonlocal Nanobeams." In *Nonlocal Mechanics Approaches for Modelling Localized Deformations: Applications to Fatigue, Fracture, and Damage in Engineering Materials and Structures,IIT, Hyderabad, Dec 14-16,2026.*

Technical Skills

Programming	MATLAB , C++, Python
Numerical methods	Element-Free Galerkin, Finite element method
Simulation tools	ANSYS
Typesetting	LaTeX, Beamer
Version control	Git / GitHub

Additional Information

Languages	English (fluent), Malayalam (native), Hindi (basic), Tamil (basic)
Interests	Classical mechanics, web designing, online chess